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# 1.0 Executive Framework & ERP Architectural Framework

Ye comprehensive operational guide **OMPI India LLP** ke proprietary Centralized ERP platform ka complete functional specification hai. OMPI India LLP Italy ke premier industrial clutch & brake manufacturer **OMPI S.r.l. (Modena, Italy)** ka authorized Indian distributor hai, jo heavy mechanical presses, forging presses, aur metal forming machines ke liye pneumatic friction clutches, hydraulic brakes, dual safety valves, aur rotating unions supply karta hai.

### On-Premise LAN Architecture & Zero Recurring Cloud SaaS Cost

OMPI India Central ERP ek high-speed, local area network (LAN) based architecture par design kiya gaya hai jo Plot W-112, MIDC Bhosari, Pune office me dedicated local server PC par run hota hai:

**1. 100% Data Sovereignty**
Entire financial, customer, and import pricing data stays on local office LAN. No public leak.

**2. Zero Recurring Fees**
Eliminates expensive monthly per-user cloud SaaS subscriptions (saving ₹1.5L–₹3L annually).

**3. Sub-5ms LAN Speed**
Instant local response even during internet downtime. Connected across office PCs and mobile tablets.

## 1.1 The 4 Core Engines & 33 Modular Structure

ERP platform 4 tightly integrated operational engines me structured hai jisme exactly 33 functional modules hain:

| Engine Name | Module Count | Operational Domain & Business Value Delivered |
|---|---|---|
| **Engine 1: Commercial Sales** QUOTATION & CRM | **10 Modules** (Mod 01 – 10) | Client master, credit profiling, OMPI SKU master, multi-currency conversion, fast-quote PDF engine, compatibility matrix, and 30/50/20 proforma invoice workflows. |
| **Engine 2: Engineering BOM** ITALY LANDED COST | **7 Modules** (Mod 11 – 17) | Multi-level clutch assembly BOM, EUR import purchase orders to OMPI Italy, customs landed cost calculator (BCD/SWS/IGST/CHA), shipment tracking, and torque sizing. |
| **Engine 3: Spares Inventory** WAREHOUSE & QC | **8 Modules** (Mod 18 – 25) | Bhosari warehouse rack/bin management, friction lining tracking, unit serialization & heat codes, inward GRN inspection, outward delivery challan, and re-order alerts. |
| **Engine 4: Financial Ledger** GST & GOVERNANCE | **8 Modules** (Mod 26 – 33) | B2B GST tax invoicing, GSTR-1 Excel exporter, 30/50/20 milestone accounts receivable, debtor aging (30/60/90 days), RBAC staff permissions, and immutable audit logs. |

## 1.2 Shared Ecosystem with Dalal Machine Tools

OMPI India LLP aur Dalal Machine Tools Agency Pvt. Ltd. ka commercial relationship software me seamlessly connected hai:

- **Clutch Retrofit Funnel:** Dalal ki website par aane wale heavy press buyers ko OMPI Italy clutch-brake safety retrofits cross-sell kiye jaate hain.
- **Shared Chakan & Bhosari Warehouses:** Inventory and delivery challans can track items moving between Bhosari central stores and Chakan machine stockyard.
- **Independent Corporate Billing:** Dono entities ke GSTIN, bank accounts, balance sheets, aur audit logs 100% isolated hain.

## 2.0 Engine 1: Commercial Quotation & Sales Pipeline

Engine 1 OMPI sales engineers ka frontline system hai jo client inquiries ko fast capture karta hai, clutch sizing verify karta hai, aur branded PDF quotations generate karta hai:

| Module Code & Name | Core Technical Function | Practical Business Use (Kaam) |
|---|---|---|
| **Module 01** Client Master & Credit Term Profiler | Buyer corporate profile, validated GSTIN, plant locations, credit limit allocation, payment terms (default 30/50/20), key contact personas. | Sales team ko client ki credit rating aur historical order volume ka instant profile milta hai; bad debts prevent karta hai. |
| **Module 02** OMPI Technical Catalog & SKU Master | Parametric database of OMPI Italy products: Pneumatic clutches, hydraulic brakes, dual safety valves, rotating unions, friction linings. | Standardized OEM product specs; sales engineer can select exact model code without manual catalogue search. |
| **Module 03** Multi-Currency Pricing Engine | EUR base catalog price, forward exchange rate entry (online fetch when connected / manual override), duty markups, company margin. | Calculates exact INR landed base price from EUR ex-factory quote, ensuring currency volatility does not erode margins. |
| **Module 04** Fast-Quote PDF Generator | 1-Click generation of formal PDF quotation with OMPI corporate header, dimensional fitment diagram, warranty terms, payment split. | Quotation turnaround time drops from 4 hours to under 2 minutes; buyer receives publication-grade proposal instantly. |
| **Module 05** RFQ Lifecycle & Status Pipeline | Visual Kanban / tabular pipeline tracking status: New RFQ → Sizing Review → Quote Sent → Negotiation → Won / Lost. | Sales manager can track all open proposals across India and prevent deal leakage with automated follow-up flags. |
| **Module 06** Machine Compatibility & Retrofit Matrix | Algorithm mapping customer press capacity (tonnage, flywheel RPM, shaft diameter) to recommended OMPI clutch model and torque. | Prevents incorrect clutch sizing by technical sales staff; ensures safety compliance and correct mechanical fitment. |
| **Module 07** Customer Discount & Margin Matrix | Tiered discount structure based on buyer classification (OEM Press Builder, Tier 1 Forging Plant, End-User Retrofit). Floor margin guard. | Protects company profitability by preventing sales reps from discounting below minimum allowable gross margins. |
| **Module 08** Proforma Invoice (PI) Generator | Generates official Proforma Invoice with 30% advance, 50% against dispatch, 20% on commissioning milestone terms and bank IFSC. | Enables customer to release mobilization advance immediately; legal commercial basis for starting procurement. |
| **Module 09** Sales Order (SO) Conversion Engine | 1-Click lock-in of approved Proforma Invoice into Confirmed Sales Order. Freezes agreed price, delivery timeline, and technical BOM. | Eliminates manual re-entry errors between sales negotiation and warehouse dispatch / procurement execution. |
| **Module 10** WhatsApp & Email Quote Dispatcher | Automated PDF dispatch via WhatsApp API and SMTP email with standardized corporate message and delivery confirmation logs. | Reaches plant procurement heads directly on mobile messaging where 80% of industrial purchasing decisions happen. |

## 2.1 Engine 2: Engineering BOM & Italy Landed Cost

Engine 2 international import procurement, customs valuation, and mechanical engineering calculations ko govern karta hai:

| Module Code & Name | Core Technical Function | Practical Business Use (Kaam) |
|---|---|---|
| **Module 11** Multi-Level Bill of Materials (BOM) | Hierarchical assembly BOM for combined clutch-brake units: Housing casting, friction discs, piston seals, springs, safety valve. | Enables OMPI engineers to break down complete units into replaceable sub-components for service and refurbishment. |
| **Module 12** Italy Import Purchase Order (PO) | Automated PO generation in EUR formatted for OMPI S.r.l. (Italy). Includes OEM part numbers, delivery incoterms (Ex-Works / FCA). | Direct standardized ordering to Italian factory without manual drafting; tracks outstanding international procurement. |
| **Module 13** Landed Cost & Customs Calculator | Algorithmic cost buildup: Ex-works EUR → Air/Sea freight → Assessable Value → BCD (7.5%) → SWS (10%) → IGST (18%) → CHA & inland transport. | Calculates exact per-unit landed cost in INR down to the rupee, ensuring complete clarity on gross margins before quoting. |
| **Module 14** Shipment & Bill of Lading (BL) Ledger | International tracking: Airway Bill (AWB) / Sea BL numbers, shipping line, customs port arrival at Nhava Sheva / Mumbai Airport, clearance alerts. | Provides real-time visibility into import pipeline so sales staff can give accurate delivery dates to Indian press customers. |
| **Module 15** Technical Drawing & GA Repository | Storage and instant lookup of General Arrangement (GA) drawings, CAD 2D/3D models, shaft fitment tolerances, keyway dimensions. | Enables immediate sharing of dimensional drawings with customer machine designers for flywheel machining verification. |
| **Module 16** Torque & Pneumatic Pressure Sizing | Interactive formula engine calculating dynamic torque (Nm), thermal dissipation (kJ/min), and required air pressure (5.5 to 6.0 bar). | Guarantees that selected clutch has sufficient safety margin (typically 1.5× to 2.0× of nominal press torque) to prevent slippage. |
| **Module 17** Import Duty & HSN Tariff Master | Statutory repository of customs tariff codes: HSN 84836090 (Clutches/shaft couplings), 84812000 (Safety valves) with statutory updates. | Ensures customs documentation conforms to Indian Customs and ICEGATE classifications, avoiding port penalties. |

**Landed Cost Formula Architecture (Module 13 Calculation Standard)**

Landed Cost = `CIF Value (Ex-Works EUR × FX Rate + Freight + Ins)` + `Basic Customs Duty (BCD @ 7.5%)` + `Social Welfare Surcharge (SWS @ 10% of BCD)` + `Port Clearance / CHA Charges` + `Inland Transport to Bhosari Yard`.

*IGST (18%) is paid at customs and tracked as Input Tax Credit (ITC) under Module 27 for complete cash flow optimization.


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## 2.2 Engine 3: Spares Inventory & Multi-Yard Hub

Engine 3 physical stock tracking, unit-level serialization, and warehouse operations ko govern karta hai:

| Module Code & Name | Core Technical Function | Practical Business Use (Kaam) |
|---|---|---|
| **Module 18** Real-Time Stock Ledger & Bin Management | Physical bin/rack locator at Bhosari central warehouse. Live quantity on hand, allocated to orders, and free stock. | Storekeeper can locate high-value clutch assemblies and friction pads in under 30 seconds; eliminates lost inventory. |
| **Module 19** Friction Disc & Lining Wear Parts Tracker | Dedicated inventory monitoring for high-turnover friction plates, sintered bronze pads, and rubber pneumatic bladders. | Ensures OMPI always maintains ready-stock emergency spares for breakdown repair of operating automotive press lines. |
| **Module 20** Serial Number & Heat Code Traceability | Captures unique Italian OEM serial numbers, manufacturing batch, and heat treatment codes for every individual unit. | Ensures 100% traceability; if an industrial press fails, OMPI can trace the exact batch and installation history. |
| **Module 21** Inward Material Inspection (GRN Engine) | Goods Receipt Note (GRN) generator. Physical damage inspection, packing list verification against Italy invoice, dimensional QC. | Prevents damaged or incomplete shipments from entering inventory; creates official record for insurance or OEM claims. |
| **Module 22** Outward Delivery Challan & Dispatch | Generates GST Rule 48 compliant Delivery Challan with serial numbers, customer machine code, vehicle number, and dispatch mode. | Legal dispatch authorization for goods moving to customer factory premises for mechanical erection and trials. |
| **Module 23** Reconditioning & Service Center Ledger | Logs maintenance, relining, piston seal replacement, and pressure test certifications conducted at Bhosari service bay. | Creates new revenue stream from clutch reconditioning; tracks service warranty history for repeat clients. |
| **Module 24** Scrap & Dead Stock Management | Segregation and accounting of worn-out friction linings, damaged packaging, obsolete revisions, and write-off ledger. | Keeps active inventory valuations clean and prevents dead stock from cluttering warehouse physical bins. |
| **Module 25** Automated Re-Order Alerts & Triggers | Calculates minimum stock thresholds based on lead time (typically 6–8 weeks from Italy) and triggers replenishment alerts. | Prevents stock-out of critical seal kits and valves; alerts procurement team before inventory reaches zero. |

## 2.3 Engine 4: Financial Ledger, GSTR-1 & Governance

Engine 4 statutory GST compliance, milestone receivables, and enterprise audit security ko govern karta hai:

| Module Code & Name | Core Technical Function | Practical Business Use (Kaam) |
|---|---|---|
| **Module 26** GST Tax Invoice Engine | B2B Tax Invoices with SAC/HSN codes, CGST/SGST (18% for Maharashtra) or IGST (18% inter-state), invoice numbering sequence. | Generates statutory tax invoices compliant with Indian GST laws ready for direct customer accounting dispatch. |
| **Module 27** GSTR-1 Reconciliation Exporter | 1-Click export of B2B invoices, credit notes, and HSN summary in GST Portal compliant CSV / JSON structure for CA review. | Reduces monthly GST filing time from days to 15 minutes; ensures zero mismatch between ERP books and GST portal. |
| **Module 28** Accounts Receivable & Milestone Tracking | Tracks payment receipts against commercial milestones: 30% advance, 50% dispatch clearance, 20% commissioning sign-off. | Gives real-time visibility into cash flow; prevents storekeeper from dispatching units before 50% milestone is cleared. |
| **Module 29** Overdue Aging & Payment Reminder Hub | 30 / 60 / 90 days debtor aging reports with automated payment reminder logs, follow-up history, and interest calculations. | Accelerates receivables collection from manufacturing clients; flags chronically delayed accounts to management. |
| **Module 30** Role-Based Access Control (RBAC) | Granular user permissions: Admin (Full), Sales (Quotes only), Storekeeper (Inventory & GRN), Accountant (Invoicing & GST). | Enforces internal controls; storekeeper cannot change prices, and sales reps cannot alter financial ledger entries. |
| **Module 31** Immutable Security Audit Log | Every invoice creation, price override, inventory adjustment, or deletion is recorded with User ID, IP address, and timestamp. | Provides complete forensic accountability for internal audits, eliminating fraud and unauthorized alterations. |
| **Module 32** On-Premise LAN Backup & Snapshot | Automated daily rolling database backups to local SSD + 1-click encrypted export to external USB storage for disaster recovery. | Ensures 100% data safety against server hardware failure or accidental corruption; restores system in under 15 minutes. |
| **Module 33** Executive MIS & Analytics Dashboard | Real-time business telemetry: Monthly gross revenue, gross profit margin by product line, open order book value, inventory valuation. | Gives designated partners high-level strategic intelligence on business performance and working capital allocation. |

# 3.0 OMPI India Digital Showcase & Lead Pipeline

OMPI India LLP ka public digital showcase Plot W-112, MIDC Bhosari, Pune credentials ko establish karta hai aur inbound press clutch retrofit inquiries ko capture karke directly ERP Engine 1 me route karta hai:

### Authorized Italian Manufacturer Representation Showcase

Public portal OMPI S.r.l. (Italy) ki authorized Indian distributorship ko highlight karta hai. Isme European CE safety conformity, dual monitored safety valves (EN 692 standards), aur pneumatic clutch retrofit engineering capabilities prominently feature hoti hain.

**1. Distributor Schema**
JSON-LD schema establishing official distributor identity for Google & AI search.

**2. MIDC Bhosari SEO**
Local B2B industrial SEO targeting Pune, Gujarat, and NCR automotive stamping press hubs.

**3. AEO FAQ Suite**
Engineered Q&A on clutch sizing, air pressure (5.5–6 bar), and flywheel retrofits.

## 3.1 The Inbound Web-to-ERP Handshake Workflow

Web portal par aane wale corporate buyer inquiries instantly ERP sales pipeline me feed hote hain:

1. **Inbound RFQ Submission on Digital Showcase**
   Plant maintenance head web form par press details enter karta hai (e.g. *"Need pneumatic clutch for 800T Knuckle Press, existing shaft 180mm"*).

2. **Automated Ingestion into ERP Engine 1 (Module 05 RFQ Pipeline)**
   Lead without manual intervention ERP system me **"New RFQ"** ke roop me appear hota hai, aur sales engineering team ko notification trigger hota hai.

3. **Automated Technical Sizing & Landed Pricing Lookup**
   Sales engineer Module 06 (Compatibility Matrix) me 800T press enter karta hai → System recommended model (e.g. OMPI Series 10.000 / 12,000 Nm) select karta hai aur Module 03 se landed INR cost fetch karta hai.

4. **1-Click Fast-Quote PDF Generation & WhatsApp Dispatch**
   Engineer Module 04 se 1-click me official quotation generate karta hai aur Module 10 ke through buyer ke WhatsApp aur email par transmit kar deta hai.

# 4.0 End-to-End User Journeys (Sales & Import Cycles)

Detailed walkthrough of daily business cycles across sales engineering and international procurement:

## Workflow 1: Customer Clutch Sizing & Commercial Fast-Quote Flow

**Scenario:** Auto forge shop in Pune requires urgent clutch retrofit for a 1000T Mechanical Press:

1. **Lead Intake & Sizing Calculation (Modules 01 & 06)**
   Sales engineer opens ERP, verifies customer GSTIN in Module 01, and inputs press specifications into Module 06 (Machine Compatibility Matrix). Sizing formula recommends OMPI Combined Clutch-Brake Unit (Rated Torque: 16,000 Nm @ 5.5 bar).

2. **Landed Cost & Margin Verification (Modules 03 & 07)**
   System computes landed cost from Italy price list (EUR 8,200 × current FX + customs duty). Module 07 applies standard OEM discount matrix and verifies that gross profit margin is ≥22%.

3. **Fast-Quote Generation & WhatsApp Dispatch (Modules 04 & 10)**
   Sales rep clicks *"Generate Official Quotation"*. High-resolution PDF with OMPI corporate header, dimensional GA drawing, and payment milestones (30/50/20) is created and sent to the client's procurement head via WhatsApp.

4. **Proforma Invoice & Sales Order Conversion (Modules 08 & 09)**
   Upon verbal agreement, system generates Proforma Invoice (PI). Once 30% mobilization advance is received, PI is locked into a Confirmed Sales Order (SO) with a single click.

## Workflow 2: Italy Import Procurement & Landed Cost Valuation Flow

**Scenario:** Placing back-to-back OEM purchase order with OMPI S.r.l. Italy:

1. **Triggering Import Purchase Order in EUR (Module 12)**
   Confirmed SO automatically prompts procurement desk to generate official Purchase Order to OMPI S.r.l. (Italy) in EUR, referencing exact factory part numbers and dispatch incoterms (FCA Italy).

2. **Shipment Tracking & Customs Port Clearance (Module 14)**
   Goods depart Milan airport/Genoa port. Logistics manager logs Airway Bill / Bill of Lading (BL) into Module 14. System tracks expected arrival date at Mumbai customs and alerts Custom House Agent (CHA).

3. **Customs Duty Assessment & Landed Valuation (Module 13)**
   Module 13 ingests customs Bill of Entry (BOE). Computes BCD (7.5%), SWS (10%), IGST (18%), and port charges, generating the definitive landed inventory valuation for the units.

## 4.1 End-to-End User Journeys (Warehouse Operations)

Detailed walkthrough of Bhosari warehouse material receipt, quality inspection, and serialized customer dispatch:

### Workflow 3: Inward Material Receipt & Serialized Storage Flow

**Scenario:** Wooden crates containing clutches arrive at Plot W-112, MIDC Bhosari central stores:

1. **Inward Material Unloading & Visual Check (Module 21)**
   Storekeeper inspects physical crates for transit damage. Opens Module 21 (GRN Engine) and matches crate packing list against Italy PO.

2. **Serial Number & Heat Code Tagging (Module 20)**
   Storekeeper records the etched Italian OEM serial numbers and metallurgy heat treatment codes for each clutch assembly into Module 20.

3. **Bin Allocation & Stock Ledger Update (Module 18)**
   Items are physically placed in designated heavy rack bins (e.g. *"Rack B2-Shelf 03"*). System updates live stock on hand, immediately showing availability in sales quote engine.

### Workflow 4: Outward Dispatch, Installation & Delivery Challan Flow

**Scenario:** Delivering clutch unit to customer press facility for mechanical erection:

1. **Milestone 2 Payment Clearance (Module 28)**
   Customer remits 50% milestone payment prior to dispatch. Accounts team verifies bank receipt and unlocks the order in Module 28.

2. **Delivery Challan Generation with Serial Numbers (Module 22)**
   Storekeeper generates official GST Rule 48 Delivery Challan. Specific serialized unit is allocated, and transporter vehicle number is logged.

3. **Onsite Erection & Power Testing (Module 23)**
   OMPI service engineer mounts clutch unit on customer press flywheel, connects 5.5 bar pneumatic air lines, tests safety dual valve timing, and issues mechanical commissioning certificate.

# 4.2 Invoicing, GST Compliance & Security Governance

Statutory B2B tax invoicing, GSTR-1 returns, and on-premise system security controls:

## Workflow 5: Tax Invoicing, GST Compliance & Debtor Settlement Flow

**Scenario:** Final commissioning completion and monthly statutory GST return filing:

1. **B2B GST Tax Invoice Generation (Module 26)**
   Upon successful press trial, Module 26 issues statutory B2B Tax Invoice. Calculates CGST + SGST (18% for Maharashtra) or IGST (18% inter-state) under HSN 84836090.

2. **Final 20% Milestone Settlement (Module 28)**
   Customer remits remaining 20% commissioning balance. System records payment against invoice, balances customer ledger, and closes the project file.

3. **Monthly GSTR-1 Excel / JSON Export (Module 27)**
   At month end, accountant clicks *"Export GSTR-1 Return"*. System generates standardized Excel file with all B2B invoices and HSN summaries ready for CA filing on the GST portal.

## 4.3 Administrative Security & SecOps Architecture

### Role-Based Access Control (RBAC)

Users are assigned distinct roles: **Admin** (Managing Partner), **Sales Engineer** (Quotes & Catalog), **Storekeeper** (Inventory & DC), and **Accountant** (Invoicing & GST). Strict permissions prevent unauthorized changes.

### Password Salting & Rate Limiting

All staff passwords are cryptographic salted hashes using **Bcrypt with 10 salt rounds**. Zero plaintext storage. API endpoints protected by express-rate-limit to prevent brute-force intrusion.

# 5.0 LAN Hardware Specs, Disaster Recovery & Governance

On-premise infrastructure requirements, disaster recovery protocols, and ongoing operational maintenance:

### Minimum Recommended Server PC Hardware

- **Processor:** Intel Core i5 / i7 (10th Gen or newer) or AMD Ryzen 5.
- **System Memory:** 16 GB DDR4 RAM (ensures instant sub-5ms caching).
- **Storage Drive:** 512 GB NVMe SSD (fast database read/write cycles).
- **Power Backup:** 1 KVA dedicated Online UPS (prevents sudden power cuts).
- **Local Network:** Gigabit Ethernet Switch (1000 Mbps) or Wi-Fi 6 router.

### 3-Tier Disaster Recovery Architecture

- **Tier 1 (Local):** Automated daily rolling database snapshots on server SSD.
- **Tier 2 (External):** Weekly encrypted export to an external USB hard drive.
- **Tier 3 (Offsite):** Monthly encrypted backup stored at secondary safe location.
- **RTO / RPO Target:** Recovery Time Objective under 30 minutes; RPO max 24 hours.

## 5.1 12-Hour Staff Training & Operational Handoff

**1. Live Interactive Training**
12 Hours dedicated hands-on training for OMPI sales, warehouse, and finance personnel.

**2. High-Def Video Library**
Recorded video walkthroughs of all 33 modules for future employee onboarding.

**3. Standard Operating SOP**
Printed operational manual for Bhosari stores desk and sales quotation desk.

## 5.2 Technical Support, In-Scope Maintenance & Documentation Governance

Ye manual OMPI India LLP ke internal staff, warehouse operators, aur commercial sales engineers ke liye definitive reference blueprint hai:

### In-Scope Implementation Support

Engagement ke 8-month period ke dauran sabhi 33 modules ke liye in-scope operational support, bug resolution, aur LAN database snapshot monitoring shamil hai.

### Commercial & Legal Governance Reference

Sabhi financial terms, payment milestone schedules (30/50/20), statutory compliance, aur IP transfer provisions formally **Proposal & Master SOW (DMTA-OMPI-SOW-2026)** ke dwara govern kiye jaate hain.